First-stage EA, Case 1

F_{Inner}

1010.0
1009.5
1009.0
1008.5
1008.0
1007.5
1007.0
1006.5

0 20 40 60 80 100

Generation

Mean of 10 runs
±1 std

First-stage EA, Case 2

F_{Inner}

1007.0
1006.5
1006.0
1005.5
1005.0
1004.5

0 20 40 60 80 100

Generation

Mean of 10 runs
±1 std

Second-stage EA, Case 1

Normalized optimality gap

1.0
0.8
0.6
0.4
0.2
0.0

0 100 200 300 400 500

Generation

Run 1
Run 2
Run 3
Run 4
Run 5
Run 6
Run 7
Run 8
Run 9
Run 10
Mean

Second-stage EA, Case 2

Normalized optimality gap

1.0
0.8
0.6
0.4
0.2
0.0

0 100 200 300 400 500

Generation

Run 1
Run 2
Run 3
Run 4
Run 5
Run 6
Run 7
Run 8
Run 9
Run 10
Mean